Computer Networks

Sample educational document for testing a PDF question-answering chatbot.

1. What is a computer network?

A computer network is a group of connected computing devices that exchange data and share resources. Networks can use wired or wireless links.

2. Network types

A Local Area Network (LAN) connects devices within a limited area such as a home, laboratory, or office. A Wide Area Network (WAN) spans a larger geographic area and can connect multiple LANs.

3. Network topologies

Topology describes how devices are arranged. Common topologies include star, bus, ring, mesh, and tree. In a star topology, devices connect to a central switch or hub.

4. TCP and IP

Internet Protocol (IP) addresses and routes packets between networks. Transmission Control Protocol (TCP) provides reliable, ordered delivery between applications by using acknowledgements and retransmission.

5. DNS

The Domain Name System (DNS) translates human-readable domain names into IP addresses. This helps users access services without memorizing numeric addresses.

Note: This is a short synthetic sample created for software testing, not an official textbook.